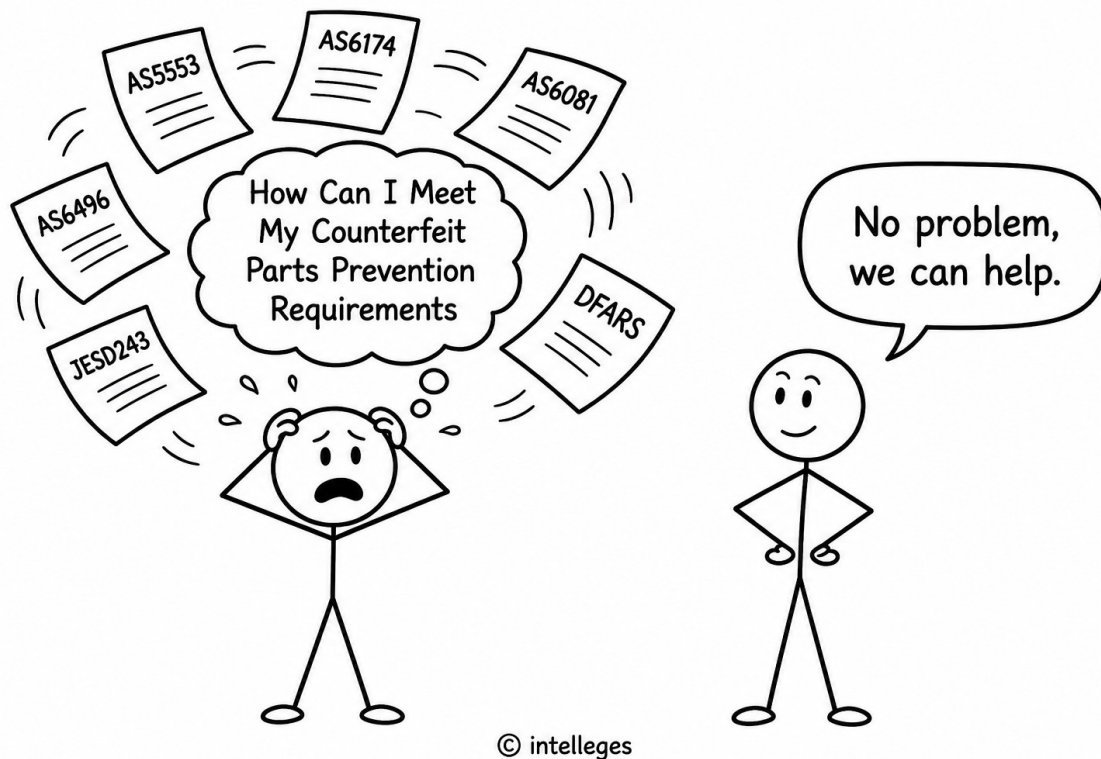


How Intelleges Helped Honeywell Aerospace Integrate Counterfeit Parts Prevention Into the Aerospace Supply Chain

From Standards Engineering to Supplier Maturity, Procurement Integration, and FCMS Execution

In 2019, Honeywell Aerospace, like many aerospace and defense contractors, faced a major challenge: how do you integrate counterfeit-parts-prevention standards into the procurement process and the supplier quality management process?



6	22	151	5
Standards	Focus Areas	Mapped Rows	Cumulative Levels

The compliance problem starts before the software.

The problem was much larger than issuing a policy or sending suppliers a questionnaire.

Counterfeit-parts requirements affect how suppliers are selected, how parts are sourced, what evidence accompanies those parts, how traceability is established, how nonconforming material is controlled, how suspect parts are investigated and reported, how distributors are qualified, what records are retained, and how procurement and supplier-quality personnel determine whether a supplier's controls are adequate.

Having worked with Intelleges since 2000, Honeywell Aerospace elected to use the Intelleges Federal Compliance Management System (FCMS) to implement the program.

But that decision hides the complexity of the overall effort.

Before FCMS could manage the compliance process, the requirements themselves first had to be established. And establishing the requirements was only the first challenge.

Before FCMS can manage a compliance program, an organization first has to determine:

- What requirements apply?
- How do those requirements relate to one another?
- Where do multiple standards address the same underlying control?
- Where do they differ?
- Which requirements apply to which types of suppliers?
- What constitutes satisfactory implementation?
- What does increasing supplier capability look like?
- What evidence demonstrates that capability?
- How should those requirements be converted into supplier controls?
- How should those controls be integrated into procurement and supplier quality management?
- How should the resulting process be deployed, monitored, reviewed, corrected, approved, evidenced, and repeated?

That was the standards-engineering and compliance-design problem Intelleges had to solve first.

The Requirements Problem

There was no single counterfeit-parts standard that could simply be converted into a questionnaire.

The relevant requirements were distributed across six separate regulatory and industry sources:

AS5553 + AS6174 + AS6081 + AS6496 + JESD243 + DFARS

These sources addressed different portions of the aerospace and electronics supply chain.

Some requirements applied primarily to manufacturers.

Others focused more heavily on distributors and brokers.

Some addressed purchasing controls.

Others addressed traceability, testing, inspection, reporting, nonconforming material, supplier approval, contractual requirements, records, packaging, returns, quality management, and other aspects of counterfeit-parts prevention.

The standards also used:

- different terminology;
- different clause numbering;
- different organizational structures;
- different levels of detail; and
- different approaches to describing similar controls.

The same underlying control might appear in several standards under entirely different clause numbers and language.

Conversely, two requirements that appeared similar might impose materially different expectations.

Before Honeywell Aerospace could reliably assess its suppliers, Intelleges therefore had to answer a much more fundamental set of questions:

- Which requirements actually applied?
- Where did the six standards overlap?
- Where did they differ?
- Which requirements were equivalent?
- Which imposed additional obligations?
- Where did one standard demand more than another?
- How should requirements applicable to different supplier types be distinguished?
- Where were there gaps?
- What constituted satisfactory implementation?
- What evidence should a supplier be expected to produce?
- How could overlapping requirements be consolidated without weakening the underlying control?
- How could all of this be translated into a repeatable procurement and supplier-quality process?

That was the real starting point.

Building a Six-Standard Counterfeit-Parts Crosswalk

Intelleges, led by Intelleges CEO John Betancourt, performed this analysis.

The substantive SAE, DFARS, standards-harmonization, and maturity-model analysis was performed by Intelleges. Honeywell Aerospace did not perform or co-develop this analytical work.

Intelleges developed a six-standard counterfeit-parts crosswalk harmonizing:

AS5553 + AS6174 + AS6081 + AS6496 + JESD243 + DFARS

The objective was to normalize requirements written in six different structures into one analytical framework in which equivalent, overlapping, complementary, and differing requirements could be examined together.

This was the difficult intellectual work that had to occur before a supplier assessment could be built.

Intelleges organized the requirements into 22 common counterfeit-parts focus areas and then mapped 151 rows into that common structure.

6 Standards	22 Focus Areas	151 Mapped Rows	5 Cumulative Levels
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This was not a checklist of six documents. It was a harmonized requirements architecture - a six-standard counterfeit-parts crosswalk.

The Crosswalk Revealed the Structure of the Risk

The crosswalk did more than organize six standards.

Its distribution also revealed where the standards themselves placed the greatest control emphasis.

The exact distribution across the 22 focus areas was:

Purchasing	34	Policy	3
Parts Traceability and Control	19	Records Retention	3
Control of Non-Conforming Parts	16	Parts Availability	2
Returns	16	Investigation	2

Supplier Approval	11	Disclosure of Non-Authorization	2
Training	8	Scrap	2
Contract Requirements	7	Quality Management System	1
Internal Audits	6	Packaging	1
Control Plan	5	Preservation	1
Reporting	5	Warranty	1
Certificates of Conformance	4	Distribution Agreements	1

The heaviest concentrations were therefore in Purchasing, with 34 mapped requirements, and Parts Traceability and Control, with 19.

At the standards level, AS6081, with 46 requirements, and AS6496, with 36, represented the largest concentrations in the crosswalk.

Standard / Area	Exact Count	Standard / Area	Exact Count
AS6081	46	Purchasing	34
AS6496	36	Parts Traceability and Control	19

That concentration is significant.

AS6081 and AS6496 address important portions of the distributor and authorized-distribution environment, while Purchasing and Traceability sit directly in the path by which suspect or counterfeit parts can enter the supply chain.

The standards architecture therefore placed substantial emphasis on questions such as:

- Where are parts being purchased?
- Is the source authorized?
- What happens when an authorized source is unavailable?
- How is supplier risk evaluated?
- How is part traceability established?
- What documentation accompanies the material?
- What happens when traceability cannot be established?
- What inspection or testing is required when sourcing risk increases?
- How are suppliers and distributors approved?
- What happens when suspect material is identified?
- How is nonconforming material controlled?
- What records must be retained?
- How are returns and scrap controlled to prevent suspect parts from re-entering the supply chain?

The crosswalk therefore did not merely catalog requirements. It exposed the structure of the counterfeit-parts risk problem itself.

But Establishing the Requirements Was Only the First Challenge

Even after the six standards had been harmonized, a critical problem remained.

Knowing that a supplier is subject to a requirement does not necessarily tell you whether the supplier has implemented that requirement effectively.

A simple Yes/No assessment can conceal enormous differences in capability.

Two suppliers might both answer:

“Yes, we have a counterfeit-parts-prevention process.”

But those suppliers might be operating at radically different levels of capability.

One supplier might have little more than an informal practice or a written policy.

Another might have:

- formally documented and implemented procedures;
- trained personnel;
- electronic records;
- risk-based sourcing controls;
- approved supplier processes;
- traceability systems;
- inspection capability;
- testing capability;
- effectiveness reviews;
- formal escalation mechanisms; and
- demonstrated evidence that the controls actually operate.

Treating those suppliers as equivalent would defeat much of the purpose of the assessment.

What does progressively better implementation of each requirement actually look like?

The Five-Level Cumulative Maturity Path

Intelleges solved that problem by building a five-level cumulative maturity path directly into the six-standard counterfeit-parts crosswalk.

This distinction is essential.

The maturity model was not a separate score applied to the crosswalk after the requirements analysis had been completed.

The maturity path was embedded in the requirements architecture itself.

For each mapped requirement, Intelleges defined progressively stronger behavioral and evidentiary states.

The general structure was:

1	Capability Absent	Required process or control does not exist.
2	Capability Developing	Process or control is being developed or implemented.
3	Documented & Implemented	Required process or control is established and operating.
4	Enhanced Maturity	Must include Level 3 plus additional validation, risk management, records, review, or control.
5	Advanced Maturity	Must include Levels 3 and 4 plus the most advanced applicable practices.

The workbook made the cumulative logic explicit:

“Must include all of 3.”

“Must include all of 3 and 4.”

This was therefore not a cosmetic score from 1 through 5. It was a cumulative capability path.

The Maturity Path Was Written Requirement by Requirement

This is one of the most important features of the methodology.

Intelleges did not create five generic maturity labels and apply the same language to all 151 rows.

The maturity path was written requirement by requirement.

That was necessary because different controls mature in different ways.

A purchasing control does not mature in the same way as a training program.

A traceability control does not mature in the same way as an audit program.

A supplier-approval process does not mature in the same way as a Counterfeit Parts Control Plan.

A testing requirement does not mature in the same way as a records-retention requirement.

A Non-Conforming Material process does not mature in the same way as a Certificates of Conformance requirement.

For each mapped requirement, Intelleges therefore had to determine:

What must actually become different in the supplier's behavior, process, controls, evidence, or technical capability before that supplier can legitimately move to the next maturity level?

That is fundamentally different from assigning an arbitrary score.

Technical Note on AS6171

AS6171 was not one of the six standards harmonized as primary sources in the crosswalk.

The six crosswalk standards were:

AS5553 + AS6174 + AS6081 + AS6496 + JESD243 + DFARS

AS6171 appeared as an additional technical standard referenced within higher-maturity criteria, particularly where advanced counterfeit-part testing and inspection capability was relevant.

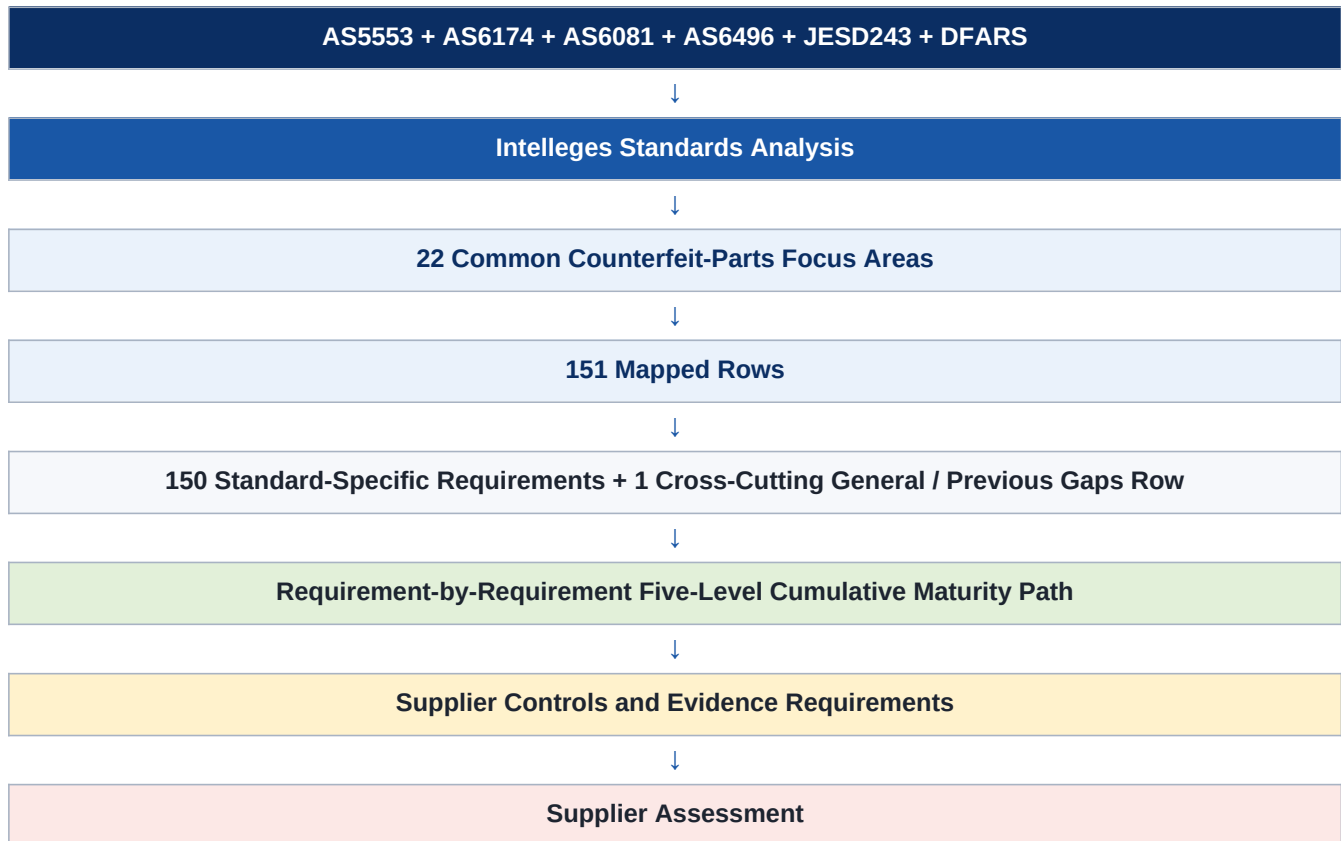
There is therefore no contradiction between describing the methodology as a six-standard crosswalk and referring to AS6171 within the maturity path.

The distinction is straightforward:

Six standards established the primary crosswalk. AS6171 was used as a technical performance reference within particular maturity anchors.

The Complete Requirements Architecture

The analytical progression was therefore:



This is why describing the work merely as a standards checklist dramatically understates what Intelleges created.

A six-standard counterfeit-parts crosswalk in which each mapped requirement had its own cumulative maturity path.

The crosswalk answered: What are the requirements, and how do six different standards relate to one another?

The maturity path answered: How does a supplier progress from inadequate capability to increasingly mature implementation of each specific requirement?

Together, the crosswalk and maturity model established the analytical foundation necessary to build an executable counterfeit-parts-prevention program.

Why This Matters

The Honeywell Aerospace experience illustrates a broader reality about compliance automation.

Software cannot solve a requirement that has not first been properly defined.

Before an aerospace and defense contractor can automate counterfeit-parts compliance, it has to understand how:

- government requirements;
- industry standards;
- contractual flowdowns;
- internal policies;
- supplier-quality requirements;
- supplier types;
- evidence requirements; and
- control expectations

fit together.

And even after those requirements have been identified, the organization still has to determine:

What does satisfactory implementation look like, and how does a supplier progress from weak or incomplete capability to increasingly mature implementation of each requirement?

That requirements-engineering work can be every bit as difficult as implementing the software that ultimately manages the process.

Bottom Line

In 2019, Honeywell Aerospace faced the same fundamental challenge confronting aerospace and defense contractors today:

How do you take complex counterfeit-parts requirements and integrate them into procurement and supplier quality management in a way that is consistent, measurable, repeatable, and auditable?

Having worked with Intelleges since 2000, Honeywell Aerospace elected to use FCMS for the implementation.

But FCMS was only part of the answer.

Before the software could manage the process, Intelleges first had to establish what the process was supposed to manage.

Led by Intelleges CEO John Betancourt, Intelleges harmonized AS5553, AS6174, AS6081, AS6496, JESD243, and DFARS into a six-standard counterfeit-parts crosswalk.

Intelleges normalized those standards into 22 common counterfeit-parts focus areas and mapped 151 rows: 150 standard-specific requirements plus one cross-cutting General / Previous Gaps row.

For each mapped requirement, Intelleges then developed a five-level cumulative maturity path, defining how supplier capability progressed from absent or developing controls through documented implementation and into increasingly sophisticated levels of risk management, evidence, testing, inspection, and control.

It was not a checklist plus software. It was a six-standard counterfeit-parts crosswalk in which each mapped requirement had its own cumulative maturity path, followed by the operationalization of that requirements architecture through FCMS.

Case study figures and counts are drawn from Intelleges standards-engineering artifacts associated with the Honeywell Aerospace CPP engagement, including All_Standards_Checklist5.xlsx.